

Marketing Mix Modelling

A Meridian MMM study with SHAP explainability, built on daily multi-territory marketing and sales data for a selected organization.

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BUSINESS OBJECTIVES

1. Quantify each channel marketing contribution,
2. Estimate ROI and marginal ROI,
3. Identify the highest-confidence path to incremental revenue.

BUSINESS CONTEXT

Why We Built This Model

Quantify Contribution

Marketing budgets are split in different channels, and a causal view is necessary to define the future choices

Map Saturation

Specific channels could have already hit diminishing returns, must be modified to optimize the budget use.

Estimate ROI & mROI

Average and marginal return by channel must be identified to have a complete view on the actual spending

Simulate Allocation

If the company had extra budget to deploy next quarter, where should it go, and how confident can the answer be?

1. Data Overview and Modelling Scope

METHODOLOGY & RATIONALE

1. Data Ingestion

- The dataset was extracted from the file conjura_mmm_data.csv, inspected in full before any filtering or modelling.

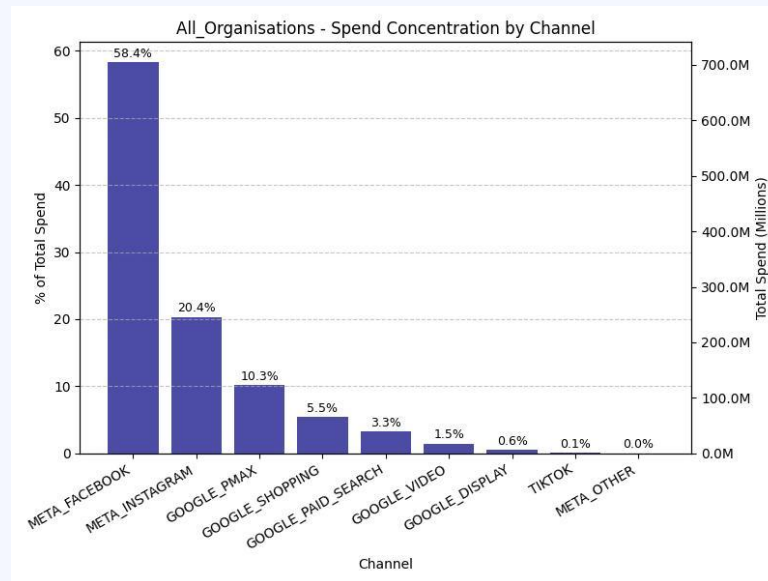
2. Data Quality

- The dataset contains multiple firms, all with different channels use and spending, for this reason most of the channels are full of missing and a complete analysis over all the dataset don't give useful information

3. Data Preparation

- The raw territory-level dataset is transformed into a Meridian-ready table by defining NET_REVENUE, as the KPI, so the model targets real economic value rather than raw purchase counts.
- A DISCOUNT_RATE control is added to capture promo-driven demand separately from media effects, and any missing media spend is filled with zero, treating gaps as inactivity rather than letting them break estimation.
- Trend, day-of-week, and month features are also added so the model can separate genuine marketing impact from seasonality and long-run growth.

SPEND CONCENTRATION BY CHANNELS: OVERALL FIRMS



2. Data Preparation and Feature Engineering – Single Organization

01 Select Organization

Best-fit organization chosen for single-firm analysis with the most reliable and informative marketing history.

02 Clean Channels

Drop not necessary variables and handle missing values, removing channels with too much missing spend data, while the remaining media gaps were filled with zero (inactive channels = zero spending).

03 Engineer Features

Removed identifiers and non-model columns, created the main KPI NET_REVENUE, and added controls such as DISCOUNT_RATE, TREND, DAY_OF_WEEK, and MONTH. Only features with both _SPEND and _CLICK columns were selected as media variables.

04 Align Geo-Time Grid

Each geo has a complete time grid aligned the dates across geographies to ensure the same time structure.

05

Chronological Split

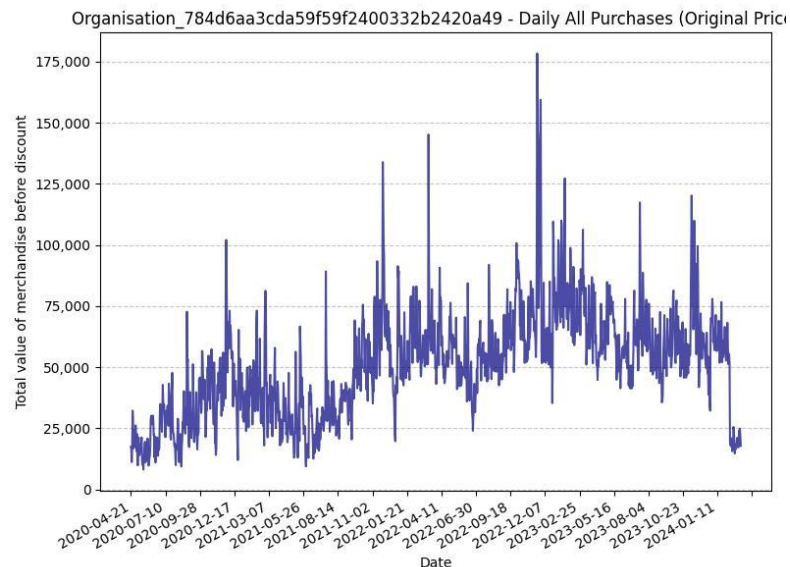
Train / validation split preserves time ordering, essential for MMM

06

Results

All these procedures, allow Meridian to separate adstock, saturation, and seasonal effects instead of confounding them.

TOTAL VALUE OF MERCHANDISE BEFORE DISCOUNT (DAILY)



Almost 3 years were considered for tracking the training data (mid 2020- mid 2023) and almost one to the test set (mid 2023-mid 2024)

3. MERIDIAN Framework

From prepared data to a trained causal MMM, in seven steps



CORE TRAINING CONFIGURATION

- **Target variable** selection: NET_REVENUE
- **Carryover:** max_lag = 8 (adstock captures delayed media effects)
- **Saturation:** Hill-type curves model diminishing returns per channel
- **Time flexibility:** knots = 50 spline knots for long-run trend
- **ROI prior:** positive LogNormal, anchored to empirical training-period ROI

REPRODUCIBILITY & VALIDATION

- **Holdout mask:** final periods per geo are marked, not split, the full-time panel stays intact
- **Fit evaluation:** R^2 , MAPE, and wMAPE vs. observed revenue
- **Reproducibility:** model saved and reloaded, no retraining on every run
- **Interpretation:** fit plot (expected vs. actual) and per-channel response curves

SPECIFIC ORGANIZATION CHOICES

MEDIA VARIABLES

META_FACEBOOK, META_INSTAGRAM
META_OTHER, GOOGLE_DISPLAY

CONTROL VARIABLES

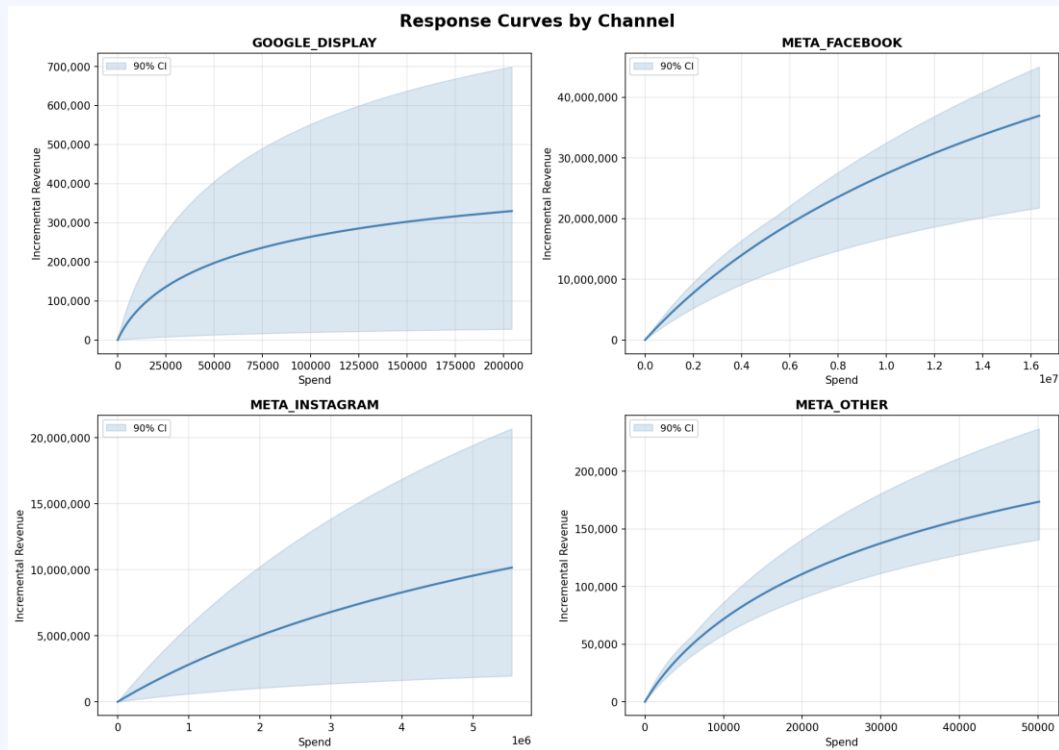
DISCOUNT_RATE (for seasonality)

TIME VARIABLES

INTERNAL KNOT (DECIDED BY
MERIDIAN ITSELF)

3. MERIDIAN Framework

MERIDIAN RESPONSE CURVES RESULTS



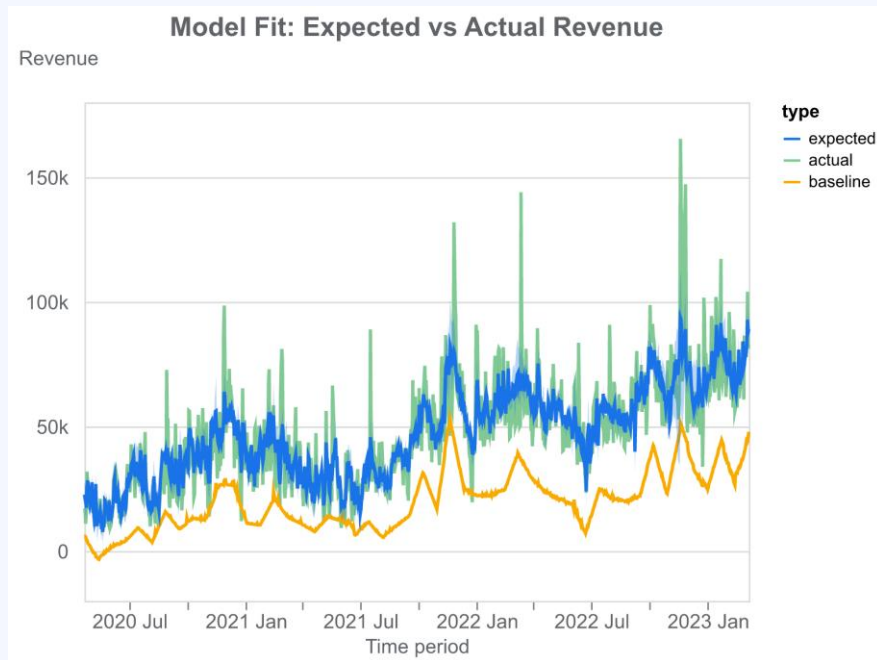
MODEL PERFORMANCE & VALIDATION

Metrics	National	GEO
R^2	0.772	0.778
MAPE	16.5%	21,8%
wMAPE	15.1%	17,9%

An R^2 near 0.77 at both geo and national level is solid enough to trust the ROI and scenario results that follow. wMAPE below 18% indicates the error is low relative to the size of the periods that matter most. Also, the curves suggest diminishing returns across all channels and the best relative efficiency appears to be Meta Facebook and Meta Instagram

3. MERIDIAN Framework

MODEL FIT RESULTS



The Meridian fit looks very solid for a business: the expected line tracks the actual revenue trend well across the full horizon, which suggests the model is capturing the **main structural drivers** of demand, including seasonality, long-term growth, and the general media effect.

The **baseline** sits clearly below both expected and actual revenue, which is what it should be seen in a properly specified MMM, because it indicates the incremental contribution from marketing is being separated from the underlying organic level.

What stands out, is that the model captures the central tendency very well but smooths some of the sharp revenue **spikes**. That is normal in MMM because peaks are often driven by short-lived promotions, stock effects, external shocks, or channel interactions that are hard to fully recover at daily granularity.

So, the results are reliable for channel contribution and budget planning but **not** intended to reproduce every **short-term** outlier exactly.

4. Channel Efficiency: ROI and Marginal ROI by Channel

All channels show $mROI < ROI$, which is the expected behaviour under diminishing returns. This means that the last euro spent on each channel generates less revenue than the historical average.



GOOGLE_DISPLAY is the only channel with mROI below 1, the clearest signal of saturation and weak marginal efficiency. This implies that each euro spent on that specific channel computes a net loss.



META_OTHER leads on both ROI and mROI, the strongest candidate for additional efficient spend, probably also because it is the lowest spending channel. **META_FACEBOOK** and **META_INSTAGRAM** stay above break-even, so both still have room to grow.

Channel	ROI (median)	mROI (median)	Status
GOOGLE_DISPLAY	2.54	0.87	Over-invested
META_FACEBOOK	3.32	2.32	Under-invested
META_INSTAGRAM	2.08	1.57	Under-invested
META_OTHER	4.97	2.59	Under-invested

An efficient reallocation would shift budget from `GOOGLE\_DISPLAY` toward the Meta channels, particularly `META\_OTHER` which shows the highest mROI (2.59) and the highest ROI (4.97), indicating it is both the most efficient channel historically and still has room for further investment.

5. SHAP – Style Contribution Analysis

WHAT DOES SHAP REALLY DO?



Channel-Level Contributions

Extracts each channel's estimated share of revenue from the fitted Meridian model



Summary & Dependence Plots

Generates visual summaries of contribution strength across channels



Geo-Specific Rankings

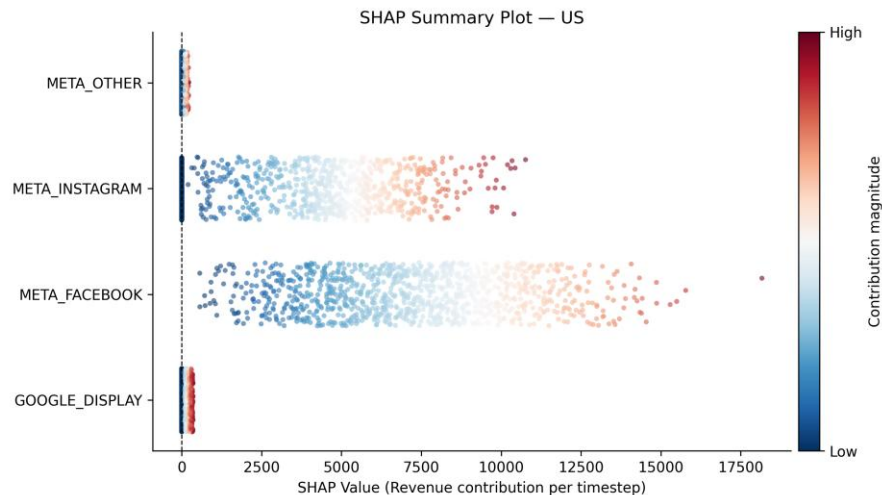
Computes importance rankings by territory, not just at the national level



Media-Intensity Comparison

Relates contribution strength back to observed spend and impression levels

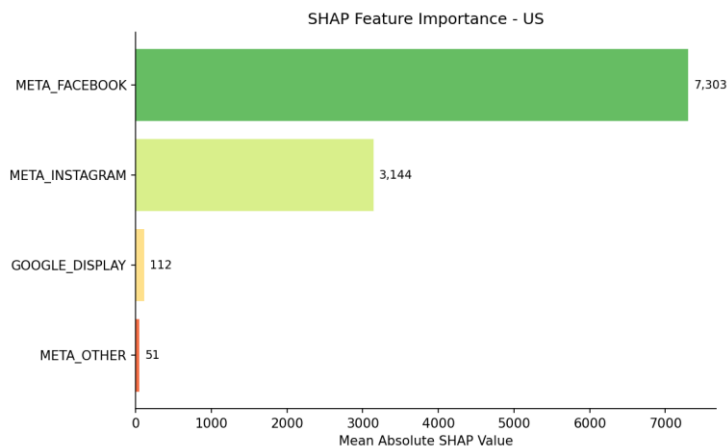
REVENUE CONTRIBUTION PER TIMESTEP BY CHANNEL - US MARKET



Meta channels contribute the most to the model's predicted outcome, showing broader distributions of positive SHAP values. Meta_Other and Google_Display contribute less overall, with values clustered closer to zero. Also, the colors suggest that higher feature values tend to drive larger (more influential) SHAP contributions.

5. SHAP – Style Contribution Analysis

CHANNEL-LEVEL IMPORTANCE BY SHAP ANALYSIS FOR US MARKET



ALERT: It must be considered that the spending of each channel is related with their importance, so an higher budget contribution (budgets below) could imply an higher SHAP result.

Also, for this reason, in the US market, **META_FACEBOOK** and **META_INSTAGRAM** show the greatest impact on the total **NET_REVENUE**.

Below the actual spending on each channel for each media channel.

TERRITORY_NAME	META_OTHER	GOOGLE_DISPLAY	META_INSTAGRAM	META_FACEBOOK
UK	\$ 18,786.06	\$ 57,980.82	\$ 1,379,157.25	\$ 4,916,769.60
US	\$ 5,927.82	\$ 55,449.27	\$ 2,052,001.72	\$ 3,776,218.90

6. Scenario Simulation: Comparing the 3 budget strategies

All scenarios share the same external extra budget (+20% of total historical spend = ~€2M) making delta_abs, incremental_purchases and incremental_roi directly comparable across rows. Scenario 3 also reallocates freed budget internally from saturated channels.

Scenario	Incr. Revenue	Incr. Purchases	Incr. ROI
1. Equal Allocation	€2,057,051	17,092	1.02
2. mROI-Optimized (all-in on META_OTHER)	€296,755	2,465	<i>~0.15</i>
3. Saturation-Aware Reallocation	€2,508,218	20,841	1.24



Best strategy: Saturation-aware reallocation wins on revenue, purchases, and ROI by combining the extra budget with an internal shift away from GOOGLE_DISPLAY.



Cautionary case: Concentrating all extra budget on META_OTHER looks very inefficient on the Meridian results, but the problem is that channel has limited historical spend and increasing the budget from 20K to over 2M saturates quickly and the results given are not reliable on the conclusion.

7. Actionable Business Implications

● Reduce

Trim exposure on GOOGLE_DISPLAY, mROI below 1 confirms it is already saturated

● Reallocate

Shift freed and incremental budget toward META_FACEBOOK and META_OTHER. The budget optimizer confirms this direction: +\$700k on META_FACEBOOK and +\$10k on META_OTHER generate an additional \$419k in revenue (+1.4%)

● Prioritize

META_OTHER first on ROI and mROI, but pace the shift to respect its scale limits

LIMITATIONS

- MAPE at the geo level (~22%) is higher than national, so territory-specific predictions carry more uncertainty than the aggregate view.
- The mROI-optimized scenario shows that efficiency metrics alone can mislead without a scalability check, the same caution likely applies to any newly scaled channel.
- ROI priors are informed by empirical training-period ROI, so results reflect recent market conditions and may shift if channel dynamics change materially.
- The budget optimizer operates under spend constraints ($\pm 50\%$ per channel), so the optimal solution is bounded by historical patterns and may underestimate gains achievable with more aggressive reallocation.

NEXT STEPS

- Pilot the saturation-aware reallocation at reduced scale before a full quarter-level shift.
- Re-run ROI, mROI, and scenario simulation on a rolling basis as new spend and revenue data arrive.
- Extend the SHAP-style explainability layer to monitor whether channel contribution rankings stay stable as budgets shift.
- Evaluate whether additional organizations warrant their own MMM once the single-firm workflow is validated in production.
- Use the optimizer output as the baseline for next quarter's budget plan: the +\$700k shift toward META_FACEBOOK and the progressive increase on META_OTHER are the two highest-confidence moves the model supports.

MARKETING ANALYTICS — PROJECT

Thank you

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